

UNDERSTANDING CRIME INFLUENCE ON REAL ESTATE USING CLOUD TECHNOLOGIES

MADE BY:-

DIVYE BEHAL



DATA COLLECTION

- Source: Kaggle
 - Link <https://www.kaggle.com/datasets/sandeep04201988/housing-price-index-using-crime-rate-data/>
- The crime data was acquired from the FBI Uniform Crime Reporting programs "Offenses Known and Clearances by Arrest" database for the year in question, held at the National Archives of Criminal Justice Data. The data was compiled and analyzed by Gabriel Dance, Tom Meagher, and Emily Hopkins of The Marshall Project; the analysis was published as Crime in Context on 18 August 2016.



WHAT IS INDEX NSA?

The US National Home Price NSA Index measures the change in the value of the US residential housing market by tracking the purchase prices of single-family homes. The index is compiled and published monthly.

GOOGLE CLOUD TECHNOLOGIES USED

- “BIG QUERY”(DATA ANALYSIS AND [VISUALIZATION) .
- “BIG QUERY ML” (TRAIN TEST & DE
THE MACHINE LEARNING MODEL)



Google Cloud

OBJECTIVE

TO SHOW THE BAD MARKET AFFECTS ON REAL ESTATES PROPERTIES IN UNITED STATES DUE TO HIGH CRIME RATES WHICH IS LEADING TO LOW "INDEX NSA" USING THE GOOGLE CLOUD TECHNOLOGY.

DATA SET

ADVANCE_DATA_PROJECT_BEHAL
merged_dataset

Initial data

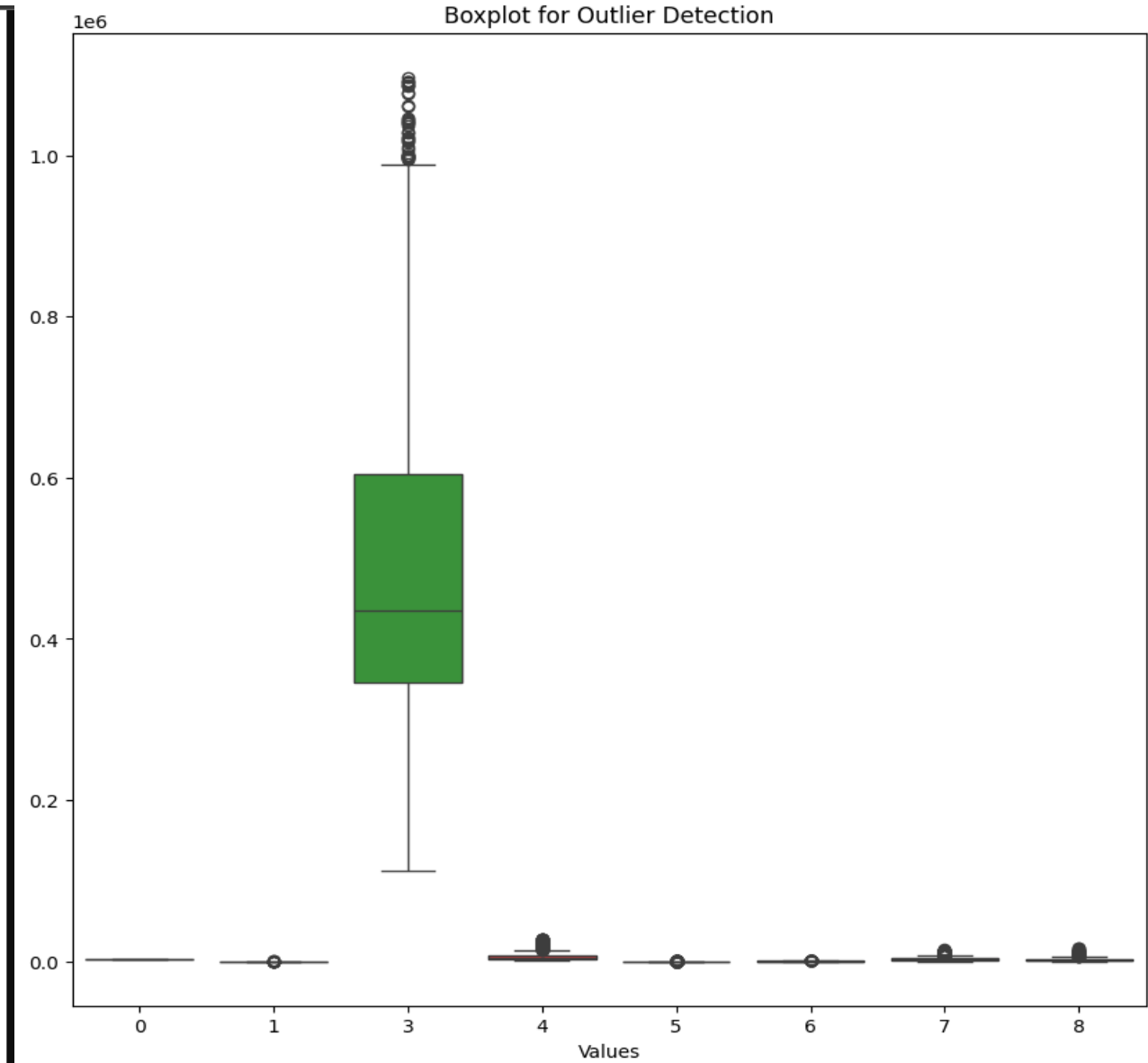
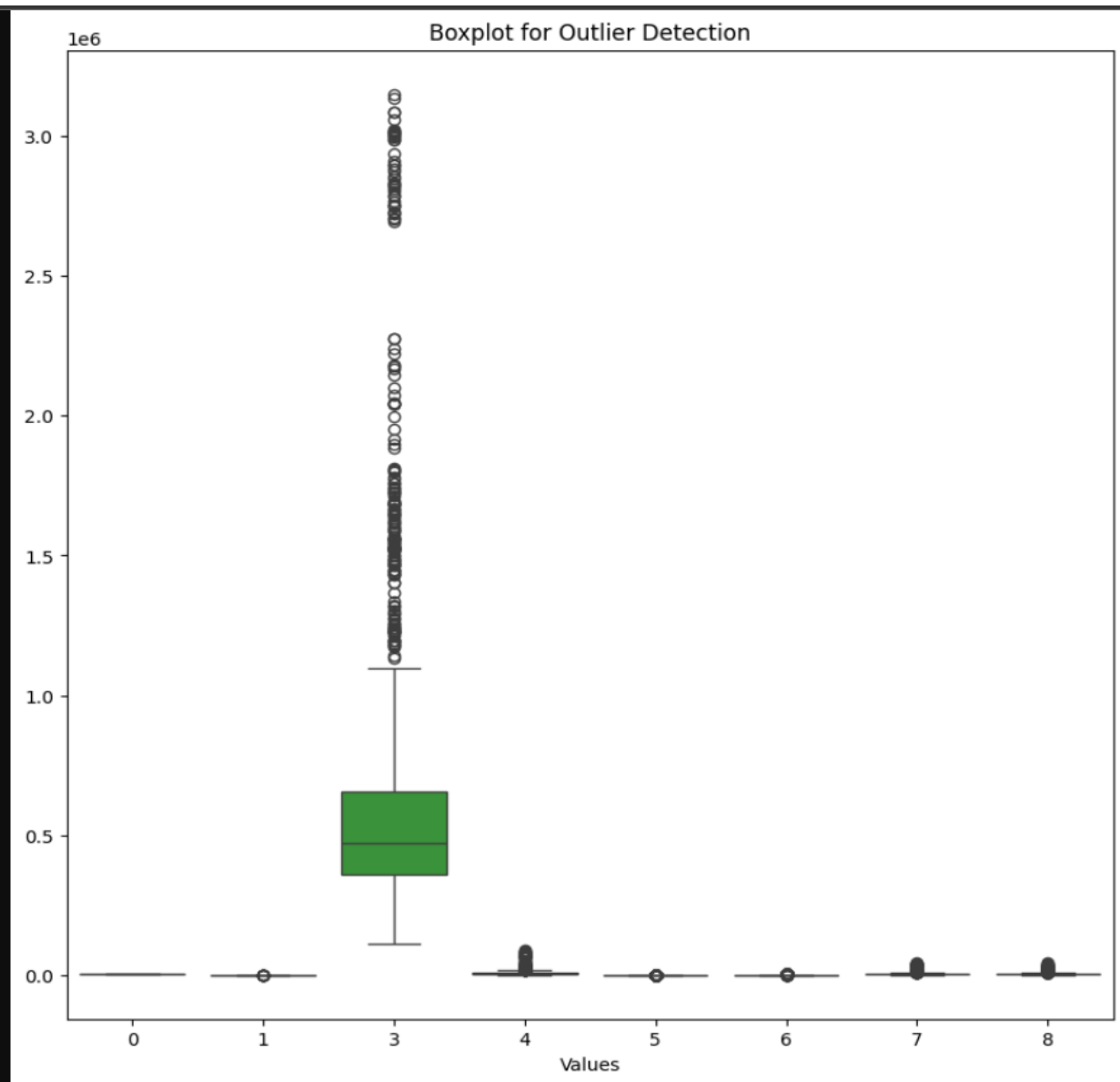
Run

Year index_nsa City, State Population Violent Crimes Homicides Rapes Assaults

1975 - 2015	20.4 - 386.2	44 Categories	112.99k - 3.15M	385 - 90.52k	1 - 960	34 - 3.75k	264 - 42.24k
1975	41.08	Atlanta, GA	490584	8033	185	443	3
1975	30.75	Chicago, IL	3150000	37160	818	1657	12
1975	36.35	Cleveland, OH	659931	10403	288	491	2
1975	20.91	Oakland, CA	337748	5900	111	316	2
1975	20.385	Seattle, WA	503500	3971	52	324	1
1975	31.215	Washington, DC	716000	12704	235	520	2
1976	40.7425	Atlanta, GA	457300	7529	154	477	3
1976	34.93	Baltimore, MD	860974	14191	200	460	5
1976	32.9175	Chicago, IL	3134499	30640	814	1179	11
1976	41.01333333	Cincinnati, OH	427045	3461	56	263	1
1976	39.99	Cleveland, OH	642298	8496	236	498	2
1976	39.21333333	Columbus, OH	595500	3287	53	328	1
1976	42.6125	Dallas, TX	888450	7244	230	591	3
1976	32.08	Aurora, CO	132000	546	2	40	1

Show / Hide data grid options 9 Columns 3,477 Rows 4 Data Types

Outliers removal



Google Cloud

sp24-41200-behal-scratch

Search (/) for resources, docs, products, and more

Search

Explorer

Viewing resources.

SHOW STARRED ONLY

- sp24-41200-behal-scratch
 - Queries
 - Notebooks
 - Shared notebooks
 - data preprocessing**
 - External connections
 - real_estate_crime_rate
 - advance_data_project_b...
 - weather_etl

data preprocessing

SAVE SHARE DOWNLOAD DELETE

+ Code + Text Commands

RAM Disk

```
[6] df = bf.read_gbq("sp24-41200-behal-scratch.real_estate_crime_rate.advance_data_project_behal") #this example uses a penguin public data:
```

```
[7] df['index_nsa'] = df['index_nsa'].round(2)
```

```
df.isna().sum()
```

Query job 6ef7e293-0736-4c2b-af01-0d5614fa0d64 is DONE. 133.9 kB processed. [Open Job](#)

Query job e8105eda-55c4-4432-91ab-810698d3f48d is DONE. 0 Bytes processed. [Open Job](#)

Query job d34273a2-6b7d-4f03-ad6a-d73617085302 is DONE. 241 Bytes processed. [Open Job](#)

Year	1736.0
index_nsa	1736.0
City_State	1736.0
Population	1763.0
Violent_Crimes	1769.0
Homicides	1768.0
Rapes	1768.0
Assaults	1769.0
Robberies	1768.0
dtype:	Float64

```
[ ] cleaned_df = cleaned_df.dropna()
```

```
cleaned_df.isna().sum()
```

```
0 0
1 0
2 0
3 0
4 0
5 0
6 0
7 0
8 0
dtype: int64
```

NULL VALUES

DUPLICATES VALUES

Type to search

Viewing resources.

SHOW STARRED ONLY

sp24-41200-behal-scratch

Queries

Notebooks

Shared notebooks

data preprocessing

External connections

real_estate_crime_rate

advance_data_project_b...

weather_etl

SUMMARY

ACTIVITY

Recent

data preprocessing

SAVE

SHARE

DOWNLOAD

DELETE

+ Code + Text ≡ Commands

RAM

Disk

⚙

✓ [129] 0s

1545	1998	127	Miami, FL	372949	9505	86	140	5482	3797	5097.21
1546	2009	203	Jacksonville, FL	810064	6772	99	218	4096	2359	1671.97
1547	1991	97	Pittsburgh, PA	372349	4294	36	300	1254	2704	2306.44

1542 rows × 10 columns

✓ 0s

cleaned_df.duplicated()

0 False

1 False

2 False

3 False

4 False

...

1543 False

1544 False

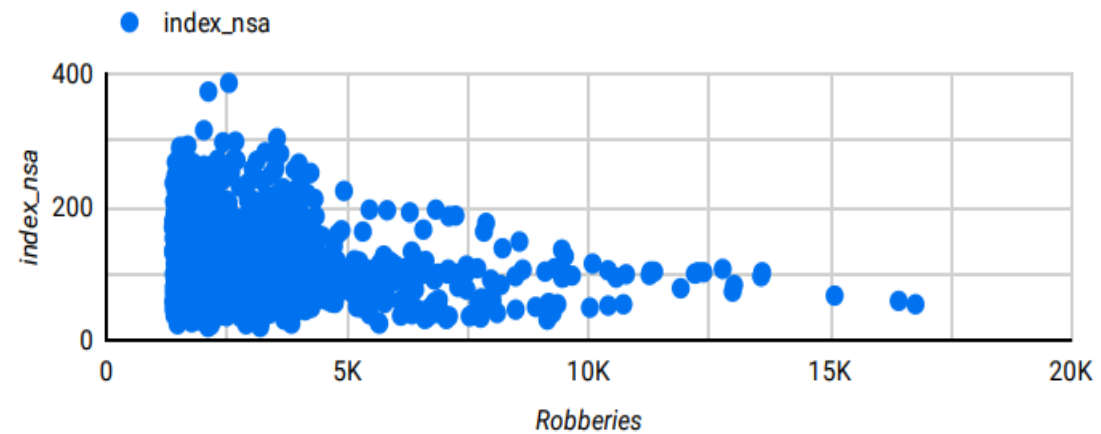
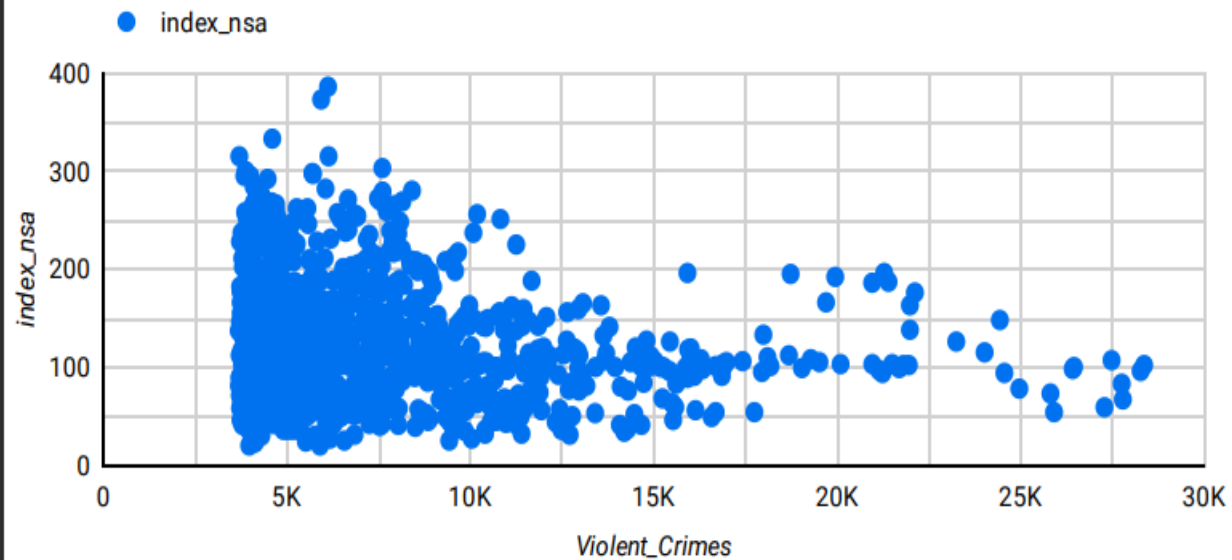
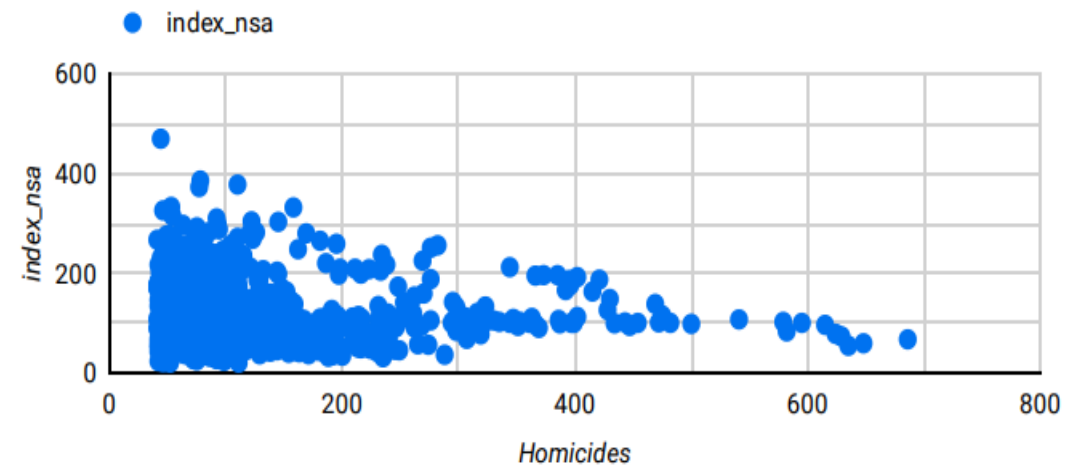
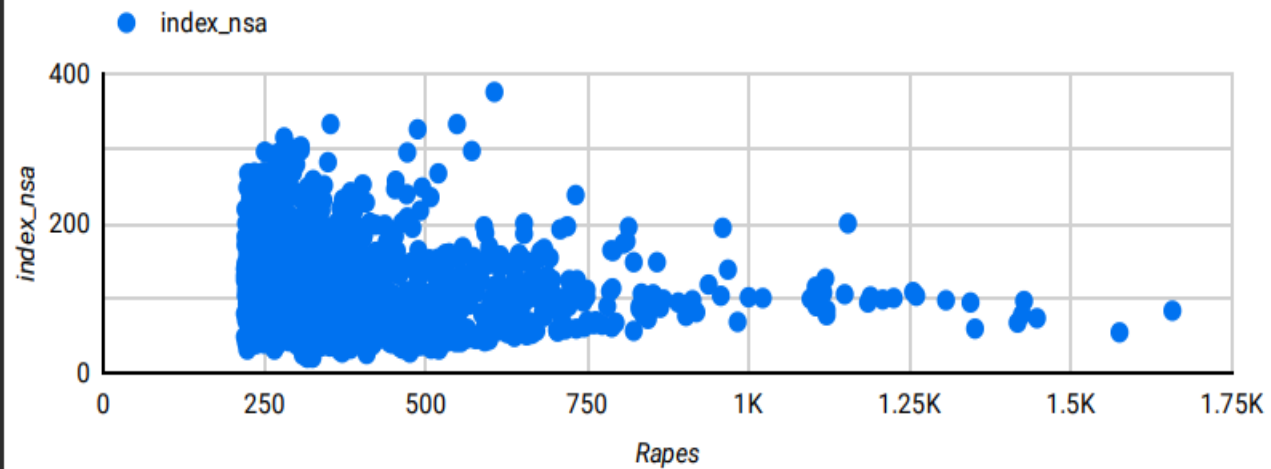
1545 False

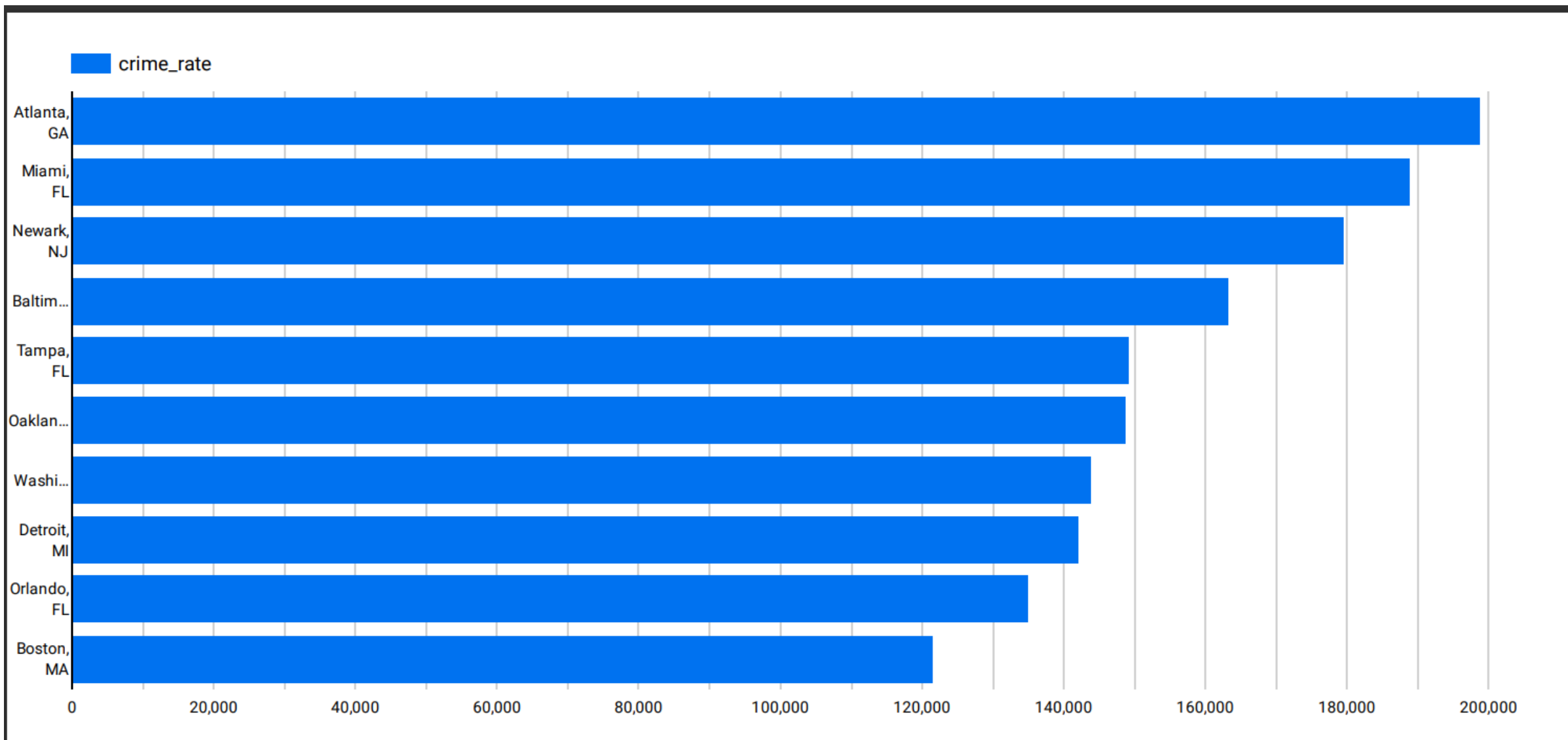
1546 False

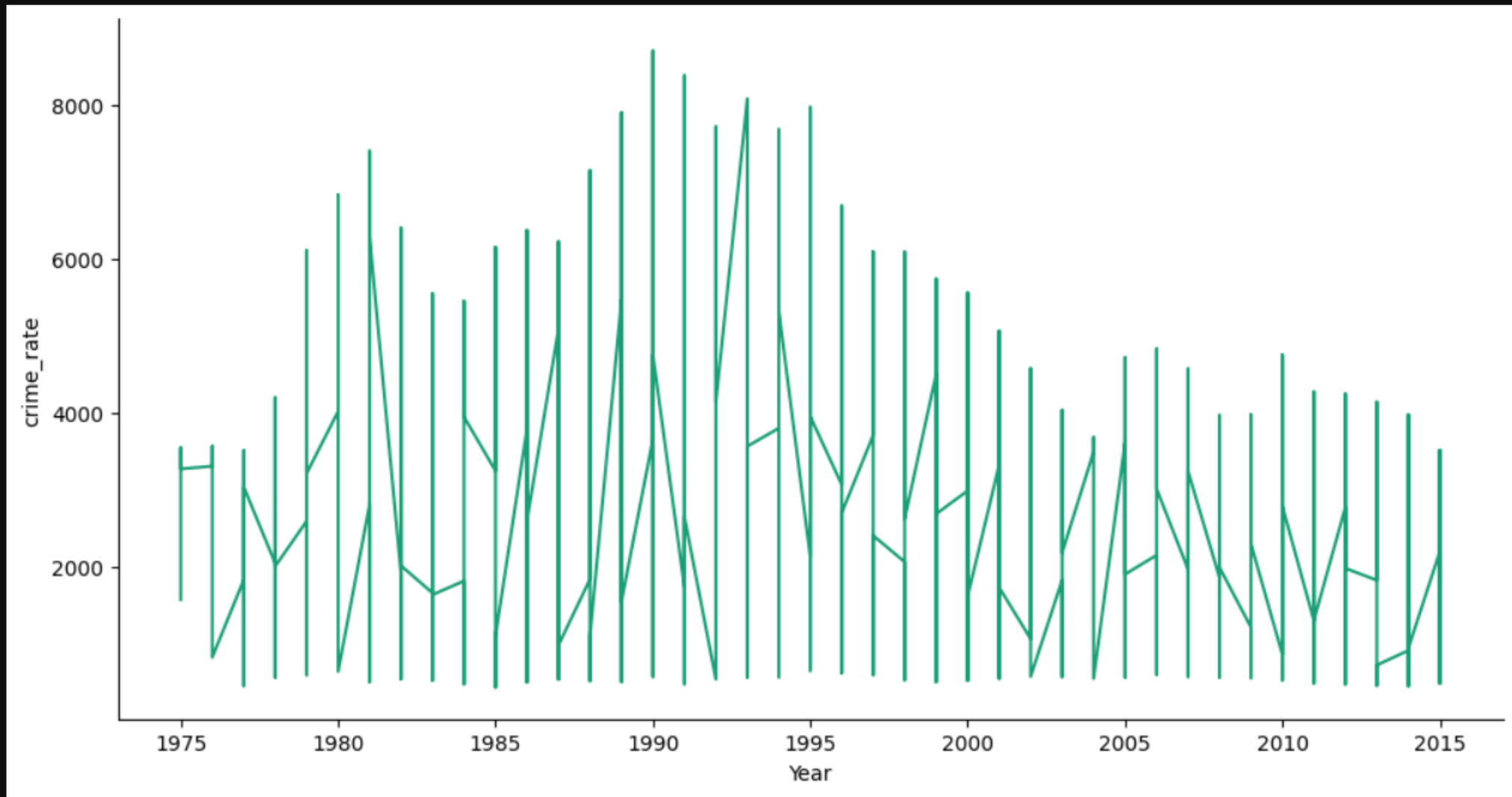
1547 False

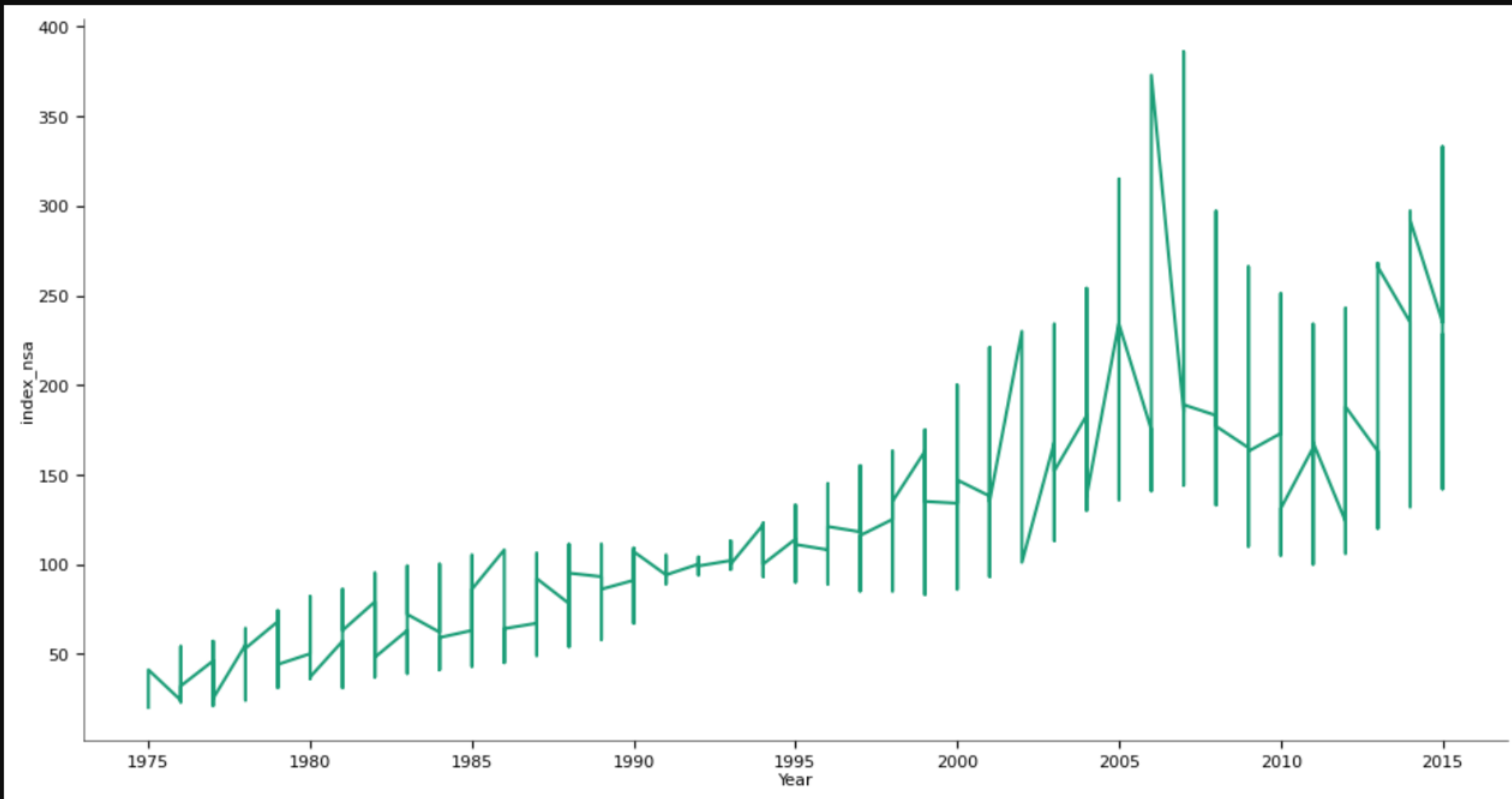
Length: 1542, dtype: bool

VISUALIZATION









TRAINING DATA (MACHINE LEARNING)

ing sp24-41... tch advance_...hal *Untitled query *Untitled query training_data *U +									
Untitled query RUN SAVE DOWNLOAD SHARE SCHEDULE MORE Query completed.									
1 SELECT 2 # features 3 Year # ex: 2015 season has March 2016 tournament games Press Alt+F1 for Accessibility Options.									
Query results SAVE RESULTSEXPLORE DATA									
JOB INFORMATION RESULTS CHART JSON EXECUTION DETAILS EXECUTION GRAPH									
Row		City_State	Population	Violent_Crimes	Homicides	Rapes	Assaults	Robberies	label
1	20	Seattle, WA	503500	3971	52	324	1492	2103	1577.36
2	36	Cleveland, OH	659931	10403	288	491	2524	7100	3152.75
3	20	Oakland, CA	337748	5900	111	316	2288	3185	3493.73
4	31	Washington, DC	716000	12704	235	520	2812	9137	3548.6
5	23	Seattle, WA	503500	4128	42	322	1601	2163	1639.72
6	25	Newark, NJ	367886	6565	99	323	2309	3834	3569.04
7	32	Washington, DC	702000	10399	188	508	2659	7044	2962.68
8	24	Oakland, CA	333100	5514	98	309	2213	2894	3310.72
9	40	Atlanta, GA	457300	7529	154	477	3518	3380	3292.81
10	34	Baltimore, MD	860974	14191	200	460	5776	7755	3296.5
11	29	Sacramento, CA	260700	2799	52	192	1368	1187	2147.3
12	30	Portland, OR	379818	4304	39	322	2030	1913	2266.35
13	32	Aurora, CO	132000	546	2	40	338	166	827.27
14	33	San Antonio, TX	150676	4187	36	322	1586	2212	1568.52

	A	B	C	D	E	F	G	H	I	J	K	
1	predicted_	Year	index_nsa	City__Stat	Population	Violent_Cr	Homicides	Rapes	Assaults	Robberies	crime_rate	
2	2272.388	1978	43	Portland, O	362497	4017	34	371	1838	1774	2216.29	
3	2065.798	1978	36	Seattle, W	489961	4900	54	360	2229	2257	2000.16	
4	3344.846	1978	27	Boston, M	616225	10035	72	475	3853	5635	3256.93	
5	3250.409	1978	35	Oakland, C	327756	5374	97	351	2152	2774	3279.27	
6	1448.545	1978	47	Austin, TX	331557	1413	35	194	549	635	852.34	
7	1884.431	1978	40	Buffalo, N	391695	3181	44	221	1195	1721	1624.22	
8	889.9368	1978	24	Honolulu, I	725259	2044	38	187	346	1473	563.66	
9	2232.184	1978	40	Sacramento	261465	3198	61	218	1337	1582	2446.22	
10	1345.465	1978	43	Tucson, AZ	309200	1654	26	155	889	584	1069.86	
11	4401.926	1978	31	Newark, N	325679	6843	109	315	2737	3682	4202.3	
12	2005.844	1978	45	Denver, CO	485547	4877	97	577	1841	2362	2008.87	
13	4023.302	1978	45	Atlanta, GA	456000	8845	144	592	3990	4119	3879.39	
14	1627.461	1978	45	Aurora, CO	131499	831	12	83	504	232	1263.89	
15	2814.336	1978	39	Washington	674000	9515	189	447	2546	6333	2823.44	
16	1359.744	1978	41	Raleigh, NC	150124	645	12	48	412	173	859.29	
17	3922.859	1978	41	Baltimore, MD	781730	14659	197	554	5811	8097	3750.4	
18	1780.086	1978	42	Albuquerque	291834	2434	37	187	1516	694	1668.07	
19	3795.925	1978	42	Miami, FL	330060	6272	96	207	3137	2832	3800.52	
20	1825.108	1978	44	Charlotte, NC	309101	2435	51	114	1561	709	1575.54	
21	1368.173	1978	46	Indianapolis	513271	3113	31	187	1114	1014	1368.17	
22	3208.62	1978	46	Orlando, FL	118393	1183	1	1	1	1	3208.62	
23	1144.359	1978	46	Columbus, OH	578338	3113	31	187	1114	1014	1144.36	
24	1903.451	1978	49	Minneapolis	374800	3748	49	49	49	49	1903.45	
25	1758.405	1978	50	Cincinnati, OH	405324	4053	50	50	50	50	1758.41	
26	3154.634	1978	50	Tampa, FL	263175	2632	50	50	50	50	3154.63	
27	1372.056	1978	51	Mesa, AZ	124937	1249	51	51	51	51	1372.06	
28	2020.270	1978	51	Cleveland, OH	614513	6145	51	51	51	51	2020.27	

MODEL PREDICTIONS

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*Untitled query X

*Untitled query X

training_data X

Untitled query

RUN

SAVE

DOWNLOAD

SHARE

```
1 SELECT
2 *
3 FROM
4 ML.PREDICT(MODEL `my_model.training_data`,
5 (
6 SELECT
7 *
8 FROM
9 `sp24-41200-behal-scratch.real_estate_crime_rate.final_crime_rate_data`
10 WHERE Year = 1978 ));
```

```
1 SELECT
2 | *
3 FROM
4 | ML.GLOBAL_EXPLAIN(MODEL `my_model.training_data`)
```

Query results

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS
Row	feature	attribution			
1	City__State	17004.54861892...			
2	Violent_Crimes	919.7459303524...			
3	Population	641.9016032253...			
4	Assaults	354.7241904126...			
5	Homicides	155.5801282537...			
6	Robberies	108.7802818559...			
7	index_nsa	56.47346305797...			

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS		EXECUTION GRAPH	
Row		predicted_crime_rate	top_feature_attribu... feature	t... attribution	baseline_prediction	prediction_value	approximation_error	Year
1	2272.388430628...		City__State	-21464.1701393...	23870.00067361...	2272.388430628...	0.0	
			Violent_Crimes	-497.532385906...				
			Population	488.5276737571...				
2	2065.798057902...		City__State	-21463.7354251...	23870.00067361...	2065.798057902...	0.0	
			Violent_Crimes	-259.433481753...				
			Assaults	-164.899857516...				
3	3344.846150017...		City__State	-21636.8262939...	23870.00067361...	3344.846150017...	0.0	
			Violent_Crimes	1125.207370824...				
			Population	-532.341761290...				
4	3250.409096210...		City__State	-20965.8103119...	23870.00067361...	3250.409096210...	0.0	
			Population	628.3073789650...				
			Assaults	-180.939637303...				
5	1448.544711699...		City__State	-21298.4431688...	23870.00067361...	1448.544711699...	0.0	
			Violent_Crimes	-1199.69495262...				

MODEL EVALUATION SCORES

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advance_...hal ▾ ×

*Untitled query ▾ ×

*Untitled query ▾ ×

training_data ▾ ×

*U > + ▾

training_data

QUERY MODEL

DELETE MODEL

EXPORT MODEL

REFRESH

DETAILS

TRAINING

EVALUATION

SCHEMA

Mean absolute error	235.7153
Mean squared error	124,680.2734
Mean squared log error	0.0483
Median absolute error	152.9471
R squared	0.9461

REFERENCES

- Effect of crimes on Housing price Index NAME: Naznee Mansoori E mail COLLEGE: IIT Kanpur
- [Meng Le Zhang](#), Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing,^{1,*} [Monsuru Adepeju](#), Conceptualization, Funding acquisition, Investigation, Writing – review & editing,² and [Rhannon Thomas](#), Data curation, Formal analysis, Investigation, Visualization, Writing – review & editing¹
- Analysis of the Correlation Between Crime Rate and Housing Price in Washington, DC, USA Based on Big Data Hanshu Yang¹, Meili Liu², Jeng-Eng Lin³, and Chun-Te Lee^{4(B)}
- Do crime hot spots affect housing prices? [Vania Ceccato](#) & [Mats Wilhelmsson](#)



**THANK YOU
ANY
QUESTIONS?**